MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics **WEEKLY CHAPTER TEST – 05 (SOLUTIONS)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 12: Surface Areas and Volumes** **MM: 20**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, factoring common terms like πr before substituting $\pi = \frac{22}{7}$ or 3.14, showing intermediate arithmetic clearly, and boxing the final answer with proper physical units ($\text{cm}^2, \text{cm}^3, \text{m}^2, \text{m}^3$).
2. Partial step marks are allocated in brackets (**[0.5 Mark]**, **[1 Mark]**, **[1.5 Marks]**, **[2 Marks]**). Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions & Assertion-Reason MATHS MANTHAN | (4 × 1 = 4 Marks)

Q1. A cylindrical bucket of height 32 cm and base radius 18 cm is filled with sand. When emptied on the ground, a conical heap of height 24 cm is formed. The radius of the base of the heap is: [1]

(A) 12 cm (B) 24 cm (C) 36 cm (D) 48 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(C) 36 cm** [0.5 Mark]

Given: Cylinder: $r_1 = 18$ cm, $h_1 = 32$ cm. Cone: $h_2 = 24$ cm, let radius $= r_2$.

Formula / Theorem Used: $\frac{1}{3}\pi r_2^2 h_2 = \pi r_1^2 h_1$

$$\frac{1}{3}r_2^2(24) = 18^2 \times 32 \implies 8r_2^2 = 18^2 \times 32 \implies r_2^2 = 18^2 \times 4 = (36)^2 \implies \boxed{\mathbf{r_2 = 36}\text{ cm}}$$

Q2. The curved surface area of a right circular cone of height 15 cm and base diameter 16 cm is: [1]

(A) 60π cm^2 (B) 68π cm^2 (C) 120π cm^2 (D) 136π cm^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(D) 136π cm^2** [0.5 Mark]

Given: $h = 15$ cm, $r = \frac{16}{2} = 8$ cm.

Formula / Theorem Used: $l = \sqrt{r^2 + h^2}$ and CSA of Cone $= \pi r l$

$l = \sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$ cm. CSA $= \pi(8)(17) = \boxed{\mathbf{136\pi}\text{ cm}^2}$.

Q3. A right triangle with perpendicular sides 3 cm and 4 cm is revolved about the side of 3 cm to form a cone. The volume of the cone so formed is: [1]

(A) 12π cm^3 (B) 15π cm^3 (C) 16π cm^3 (D) 20π cm^3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(C) 16π cm^3** [0.5 Mark]

Given: Revolving about the 3 cm side makes height $h = 3$ cm and base radius $r = 4$ cm.

Formula / Theorem Used: Volume of Cone $= \frac{1}{3}\pi r^2 h$

$$V = \frac{1}{3}\pi(4)^2(3) = \boxed{\mathbf{16\pi}\text{ cm}^3}$$

Q4. **Assertion (A):** If the radius of the base of a right circular cylinder is halved and its height is doubled, its volume is halved.
Reason (R): If the radius of the base of a right circular cone of volume V is halved and its height is doubled, the volume of the new cone becomes $\frac{V}{2}$. [1]

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: **(B)** [0.5 Mark]
Verification: For cylinder: $V' = \pi\left(\frac{r}{2}\right)^2(2h) = \frac{1}{2}\pi r^2 h = \frac{V_{\text{cyl}}}{2}$ (Assertion (A) is true).
For cone: $V'_{\text{cone}} = \frac{1}{3}\pi\left(\frac{r}{2}\right)^2(2h) = \frac{1}{2}V$ (Reason (R) is also true). Since (R) states the property of a cone rather than explaining why a cylinder's volume halves, (R) is not the correct explanation of (A).

SECTION B: Short Answer Questions MATHS MANTHAN | (2 × 2 = 4 Marks)

Q5. A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its surface area. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]
Step 1: Dimensions & Formula Declaration [1 Mark]
Radius $r = \frac{5}{2} = 2.5$ mm. Length of cylindrical portion $h = 14 - 2(2.5) = 14 - 5 = 9$ mm.
Formula / Theorem Used: $\text{SA} = 2\pi rh + 2(2\pi r^2) = 2\pi r(h + 2r)$
Step 2: Substitution & Arithmetic [1 Mark]

$$\text{SA} = 2 \times \frac{22}{7} \times \frac{5}{2} \times (9 + 5) = 5 \times \frac{22}{7} \times 14 = 5 \times 22 \times 2 = \boxed{\mathbf{220}\ \text{mm}^2}$$

Q6. Mayank made a bird-bath for his garden in the shape of a cylinder with a hemispherical depression at one end. The height of the cylinder is 1.45 m and its radius is 30 cm. Find the total surface area of the bird-bath. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]
Step 1: Converting Units & Formula [1 Mark]
$h = 1.45$ m $= 145$ cm, $r = 30$ cm.
Formula / Theorem Used: $\text{TSA} = \text{CSA}_{\text{cyl}} + \text{CSA}_{\text{hemi}} = 2\pi rh + 2\pi r^2 = 2\pi r(h + r)$
Step 2: Substitution & Answer [1 Mark]

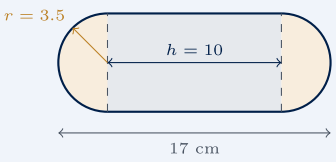
$$\text{TSA} = 2 \times \frac{22}{7} \times 30 \times (145 + 30) = 2 \times \frac{22}{7} \times 30 \times 175 = 44 \times 30 \times 25 = \boxed{\mathbf{33000}\ \text{cm}^2\ (= \mathbf{3.3}\ \text{m}^2)}$$

Q7. A wooden article was made by scooping out a hemisphere from each end of a solid cylinder. If the height of the cylinder is 10 cm and its base is of radius 3.5 cm, find the total surface area of the article. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Dimensions & Surface Decomposition [1 Mark]

Cylinder: $h = 10$ cm, $r = 3.5$ cm $= \frac{7}{2}$ cm. Two hemispherical cavities share radius $r = 3.5$ cm.



Formula / Theorem Used: $\text{TSA} = \text{CSA}_{\text{cyl}} + 2 \times \text{CSA}_{\text{hemi}} = 2\pi rh + 4\pi r^2 = 2\pi r(h + 2r)$

Step 2: Substitution & Answer [2 Marks]

$$\text{TSA} = 2 \times \frac{22}{7} \times \frac{7}{2} \times (10 + 2 \times 3.5)$$

$$= 22 \times (10 + 7) = 22 \times 17 = \boxed{\mathbf{374}\ \text{cm}^2}$$

Q8. A cone of maximum size is carved out from a solid wooden cube of edge 14 cm. Find the surface area of the cone and the surface area of the remaining solid left out after the cone is carved out. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Cone Dimensions & Slant Height [1 Mark]

Edge $a = 14$ cm $\implies$ max cone has $r = \frac{14}{2} = 7$ cm, $h = 14$ cm.

$l = \sqrt{r^2 + h^2} = \sqrt{7^2 + 14^2} = \sqrt{49 + 196} = \sqrt{245} = 7\sqrt{5}$ cm $\approx 7(2.236) = 15.65$ cm.

Step 2: Total Surface Area of Cone [1 Mark]

Formula / Theorem Used: $\text{TSA}_{\text{cone}} = \pi r(l + r)$ and $\text{TSA}_{\text{remaining}} = 6a^2 - \pi r^2 + \pi rl$

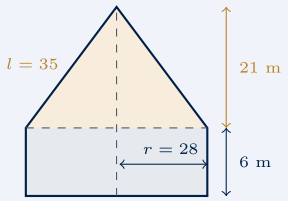
$$\text{TSA}_{\text{cone}} = \frac{22}{7} \times 7 \times (7\sqrt{5} + 7) = 154(\sqrt{5} + 1)\ \text{cm}^2 \approx 22 \times 22.652 = \boxed{\mathbf{498.35}\ \text{cm}^2}$$

Step 3: Surface Area of Remaining Solid [1 Mark]

$$\text{TSA}_{\text{rem}} = 6(14)^2 - \frac{22}{7}(7)^2 + 154\sqrt{5} = 1176 - 154 + 344.35 = \boxed{(\mathbf{1022 + 154\sqrt{5}})\ \text{cm}^2\ (\approx \mathbf{1366.35}\ \text{cm}^2)}$$

Q9. A circus tent is in the shape of a cylinder surmounted by a conical top of the same diameter. Their common diameter is 56 m, the height of the cylindrical part is 6 m, and the total height of the tent above the ground is 27 m.
(i) Find the slant height of the conical portion.
(ii) Find the total area of the canvas fabric used in making the tent.
(iii) Find the cost of canvas bought at the rate of Rs. 200 per m^2, if 30 m^2 canvas was wasted in stitching. **[6]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 6 Marks]



Given: Common radius $r = \frac{56}{2} = 28$ m; cylinder height $h_{\text{cyl}} = 6$ m; cone height $h_{\text{cone}} = 27 - 6 = 21$ m.

Step 1: (i) Slant Height of Conical Portion [2 Marks]

Formula / Theorem Used: $l = \sqrt{r^2 + h_{\text{cone}}^2}$ and Canvas $= 2\pi r h_{\text{cyl}} + \pi r l = \pi r(2h_{\text{cyl}} + l)$

$$l = \sqrt{28^2 + 21^2} = \sqrt{784 + 441} = \sqrt{1225} = \boxed{\mathbf{35}\text{ m}}$$

Step 2: (ii) Total Canvas Area Used in Tent [2 Marks]

$$\text{Canvas Area} = \frac{22}{7} \times 28 \times (2 \times 6 + 35)$$

$$= 88 \times (12 + 35) = 88 \times 47 = \boxed{\mathbf{4136}\ m^2}$$

Step 3: (iii) Cost With Wastage (30 m^2) [2 Marks]

Total canvas bought $= 4136 + 30 = 4166$ m^2.

$$\text{Total Cost} = 4166 \times 200 = \boxed{\text{Rs. } \mathbf{8,33,200}}$$

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